

Taking to the skies

Birdlike tail

Stringfellow's flying machine, 1848

POWERED FLIGHT

In 1847, the English inventors John Stringfellow and William Henson built a model airplane powered by tiny steam engines, but it failed to fully take flight. Stringfellow then built a half-size version and, in 1848, it made the first ever powered flight.

Over the centuries, people have attempted to fly by strapping on wings and lunging into the air to try to flap like a bird, but their efforts often ended in tragedy. The breakthrough for winged aircraft came when the British engineer Sir George Cayley recognized the forces that act on a plane in flight, including “lift.” It was not until 1903, however, that the Wright brothers (see pp.118–119) mastered controlled, powered flight.

DISCOVERING LIFT

Although people flew balloons in the early 1800s, British engineer Sir George Cayley believed wings were the future of flight. He experimented with kites and developed theories about wing shapes. He also built full-size gliders. In 1849, a ten-year-old boy was lifted into the air in one of Cayley's gliders—this was the first-ever manned airplane flight.

Tail and wings were made of linen stretched over a frame made of cane.



Cayley's glider

FLYING BOATS

The Wright brothers' breakthrough flight in 1903 led to the rapid development of planes. The age of air travel began in the 1930s with giant “flying boats,” such as the Short Empire, Martin M-130, and Boeing 314. In many places around the world, there were no airports yet built, but most cities have rivers. So these planes were designed to take off and land on water. The famous American “Clipper” flying boats were huge and carried passengers like they were on luxury liners.